Lab 5: Multistage amplifiers

Course	ECED 3202
Document Title	Lab 5: Multistage amplifiers
Date Issued	Oct 31, 2025
Date Due	Nov 14, 2025, 11:59pm

Relevant Lectures & Examples:

The following material is relevant to this lab:

- Lecture 7 CD and CC
- Lecture 7 Examples
- All of Lecture 10
- Lecture 10 examples

1 Purpose/Scope

The purpose of this lab is to:

- Design a multi-stage amplifier that can simultaneously achieve a high input resistance, a low output resistance and a high gain
- Make appropriate use of CE and CC stages to meet requirements
- Use a CC voltage buffer to increase input resistance seen looking into a load

2 Common emitter amplifier

You are part of a startup company developing a wireless video projector. You oversee the team designing the amplifier for the audio subsystem and are asked to design a small power amplifier to drive an $8\,\Omega$ speaker located in the bottom of the device with at least 10mW of power with <5% THD and a minimum voltage gain of 15x. Your boss wants the design on his desk by Friday, so you blow the dust off your ECED3202 notes from undergrad and get to work designing…

Requirement identifier	Requirement Description	Requirement marks
R5-01	The circuit shall be powered by a 5V supply	0
R5-02	The circuit shall drive an $8\,\Omega$ speaker load coupled to the amplifier through a $100\,\mu F$ capacitor	0

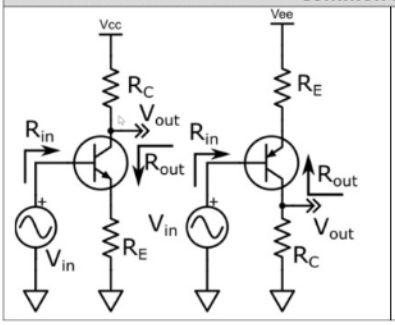
- The operating frequency range shall be 500Hz to 5KHz
- Unless otherwise noted, requirements are specified with an $8\,\Omega$ load connected to the output through a $100\,\mu F$ capacitor

Requirement Identifier	Requirement Description	Requirement points
R5-03	The input impedance of the amplifier shall be ¿ $80\,k\,\Omega$ between 500Hz and 5KHz	5
R5-04	The amplifier shall produce at least 10mW of output power with a THD < 5% at 1000Hz	5
R5-05	All transistors in the circuit shall operate within their safe operating range for current and power dissipation for inputs $\leq 100\,mVp$ (note that the response won't be linear with this amount of input signal, but that's ok)	5
R5-06	The gain of the amplifier shall be ¿ 15x (23.5dB) over the frequency range from 500Hz to 5KHz	5

3 Design and Simulation Tips

Useful equations

Common Emitter (Emitter Degeneration)

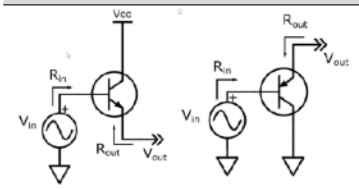


$$R_{out} = (1 + g_m R_E) r_o$$

$$R_{in} = r_\pi + (\beta + 1) R_E$$

$$A_v \approx \frac{-g_m R_C}{1 + g_m R_E} \quad (for\ r_o = \infty)$$

Common Collector



Assumes a load resistance R_E (not shown) at the emitter

$$R_{out} = r_o || \left[r_e + \frac{R_{sig}}{\beta + 1} \right] \approx r_e + \frac{R_{sig}}{\beta}$$

$$A_v = \frac{R_E || r_o}{r_e + (R_E || r_o)}$$

$$R_{in} = r_\pi + (\beta + 1) R_E || r_o$$

For the purposes of circuit design, we can simplify these equations by making the assumptions $r_o = \infty$ and $\beta + 1 \approx \beta$. This will produce some small errors, but we can address those by tweaking component values once we have a working simulation.

Using these assumptions we can derive the following equations which we'll use in designing the circuit:

For CE amplifiers:

$$R_i \approx r_\pi \left(1 + g_m R_E\right)$$

$$A_v = \frac{-g_m R_C}{1 + g_m R_E}$$

Where R_C is all the small-signal resistance seen looking out from the collector

For CC amplifiers:

$$A_v \approx 1$$

$$R_i \approx r_\pi + \beta R_E$$

Where R_E is all the resistance seen looking out from the emitter.

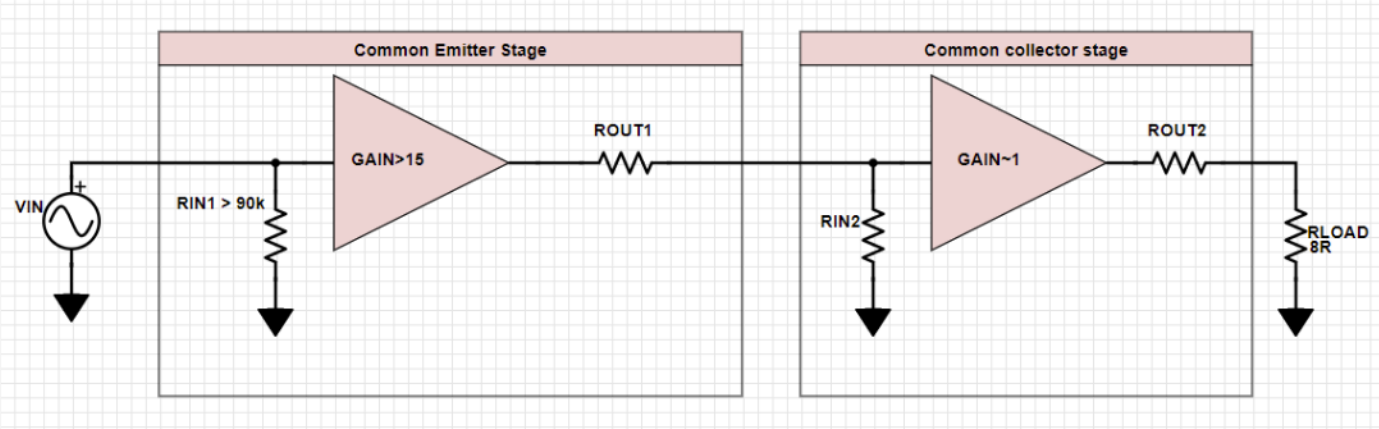
4 Designing the test circuit

In determining what topology to adopt we need to consider the following:

- The very low load impedance presented by the $8\,\Omega$ speaker is going to require a low output resistance from the amplifier to get any significant power out of it. This is a clue that you'll need a common collector (emitter follower) or common drain (source follower) stage at the output.
- The output resistor will need to be able to handle fairly large emitter/source currents. The emitter current will include both the output current and a bias current and so it must be bigger than the output current which itself has to be at least $i_{out} = \sqrt{2 \times 10\,mW / 8\,\Omega} = 50\,mAp$. You will need to select an output transistor capable of handling this much current in addition to the DC offset current.
- The amplifier also requires significant voltage gain of >15x and high input resistance of $\approx 80\,k\,\Omega$. The high gain and high input impedance tell you that you need a common emitter or common source stage as the first stage in the circuit
- So your amplifier will need at least two stages, and possibly more if we can't meet the requirements with just two.

I'll walk you through one possible solution involving BJTs (so CE and CC stages), but feel free to experiment with other topologies and transistor types. Any solution that meets the requirements is a good solution. Just keep in mind that we'll be building this amp in the next lab, so, while you don't have to stick to transistors that are in your kit, doing so will save you design work for next week.

The figure below shows how CE and CC stages can be chained together to simultaneously provide high voltage gain, high input resistance and low output resistance.



Selecting a topology and transistors:

The first thing we need to do is to pick transistors. In a power amplifier like this, the last transistor in the circuit (i.e. the output transistor) is a power transistor and will usually be bigger than the others, with the ability to handle the current needed to drive the load and to dissipate the heat associated with that current. Scanning the list of transistor models built into LTSpice, we see that that the 2N2222 can handle 800mA of DC current while providing a decently large β=200. The current handling capability is listed in the Ic[mA] column in LTSpice and can also be found in the transistor datasheet. Conveniently, this transistor is also in your kit, so when we go to build this circuit next week you'll have what you need. While it doesn't matter for this week, it's important for next week that you use the metal can 2N2222s because they are capable of dissipating more heat than the plastic case ones.

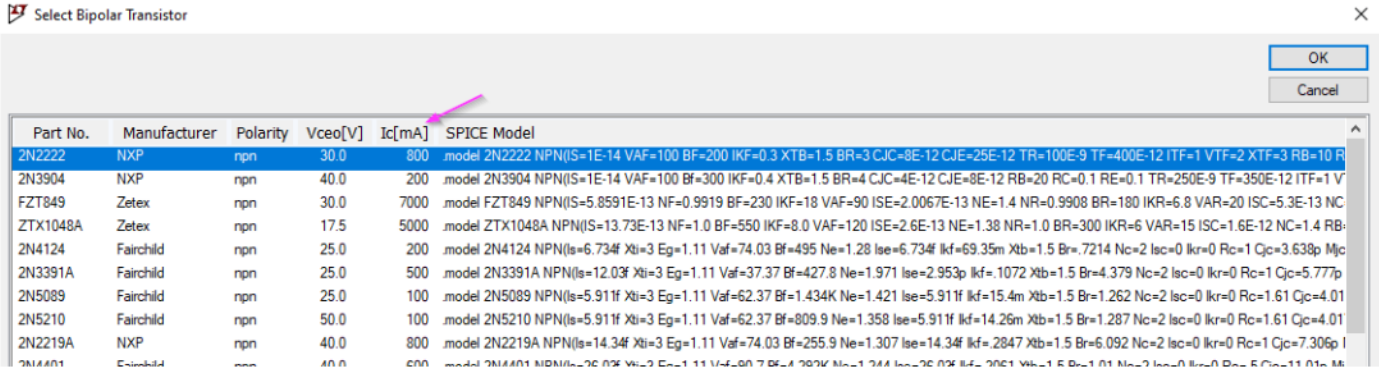


Figure 1: Current handling ability of the transistors in LTSPice

In the CE stage we don't need to handle as much current and so the current-handling specifications are not important for that transistor.

Circuit topology

We'll solve this problem using a PNP CE stage and an NPN CC stage as shown below. If you are looking for a challenge, you can try replacing the PNP CE stage with an NPN CE stage and working through the design consideration for this new design. For maximum design flexibility we include room for an emitter resistor for Q1 and we provide an emitter resistor for Q2 as well to handle the DC bias current. Without it, there would be nowhere for the bias current to go and the circuit would not function.

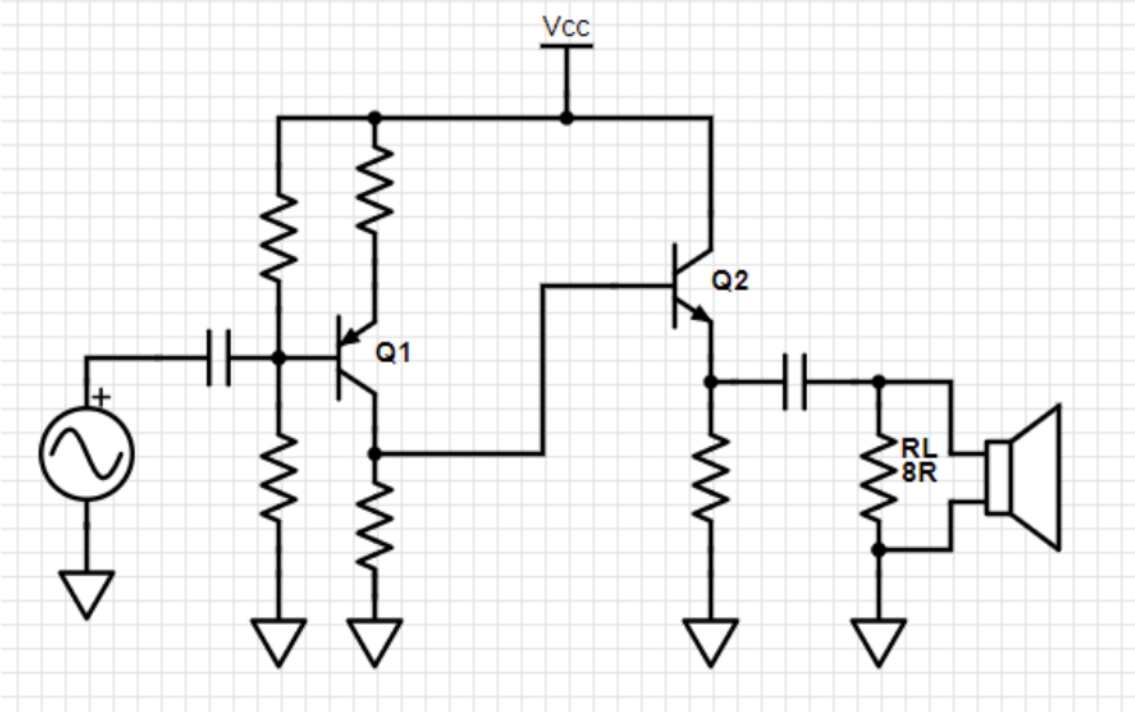


Figure 2: A possible circuit topology for solving this lab

For the PNP transistor we'll use the 2N3906 with $\beta = 200$, because it is already in your kit.

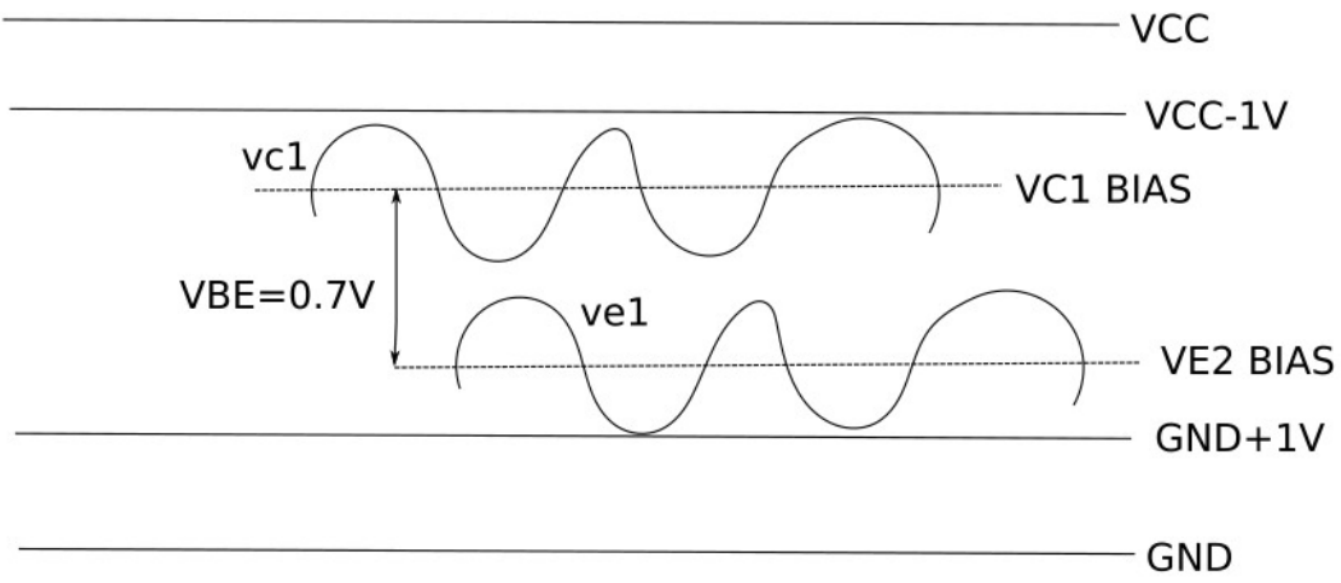
Head room and signal swing

Usually, a primary goal in amplifier design is to maximize signal swing for an AC signal within the available supply range.

Since $A_v \approx 1$ for the CC, the AC signal component at C1/B2 and E2 will be the same size, but because of the DC offsets, the available range for the signal to swing will not be the same at these two nodes. There will be a DC $V_{BE2} \approx 0.7\,V$ drop between the two nodes.

Further, as the signal approaches either the VCC rail or GND, it will start to distort. To get a "clean", undistorted signal, it's important to provide a buffer from the supply rails. As a rule of thumb, you want all your signals to stay at least 1V from either rail.

As a result, we expect our signals to look like this:



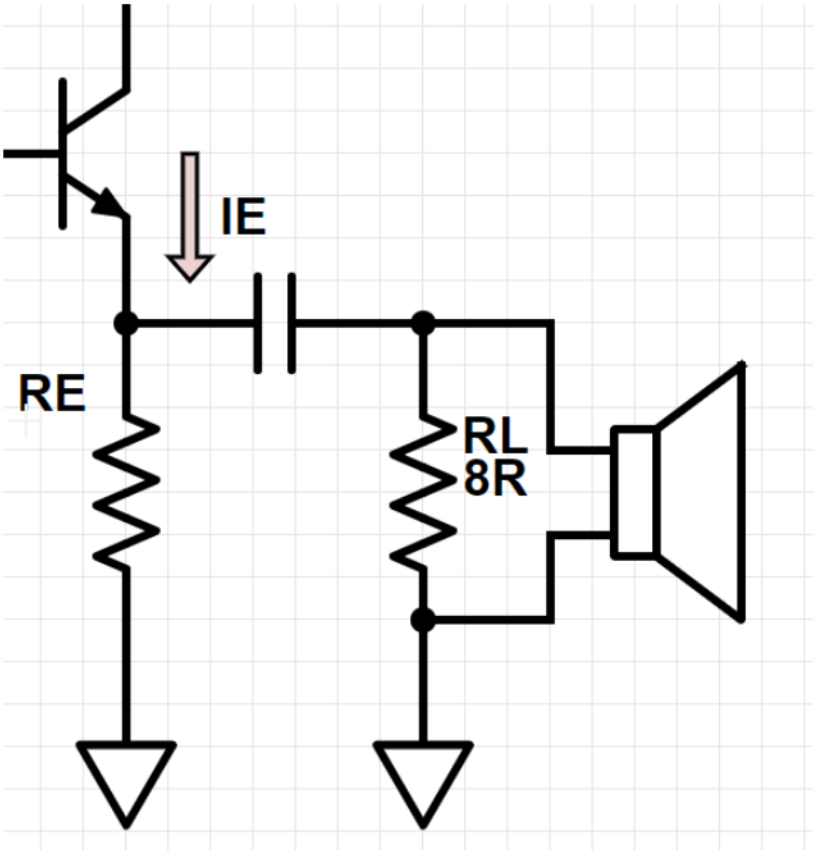
From the diagram you can see that the signal swing is maximized if the V_{BE} drop is split equally around the mid-point of $\frac{V_{CC}}{2}=2.5\,V$. With VCC=5V, this gives $V_{E2}=2.5\,V-\frac{0.7\,V}{2}=2.15\,V$ and $V_{C1}=2.5\,V+\frac{0.7}{2}=2.85\,V$. The maximum signal swing to avoid hitting the ends of the range is $\pm 1.15\,V$ since $4\,V-2.85\,V=1.15\,V$ and $2.15\,V-1\,V=1.15\,V$

Coupling capacitors:

As in previous labs, the two coupling capacitors are there to avoid having either the source or the load affect the DC bias point. $100\,\mu F$ capacitors are large enough that they present negligible impedance in the operating frequency range of 500Hz to 8 KHz while having an infinite impedance at DC.

Designing the CC stage:

In figuring out our biasing we can start with the CC stage. To get 10mW output we'll need to supply at least $I_p=\sqrt{2\times 10\,mW/R_L}=50\,mAp$. R5-02 tells us that the load is AC coupled through a $100\,\mu F$ capacitor which means that the DC bias current can't flow through the load. We need to connect another resistor R_{E2} to the emitter of the CC stage to handle the bias current $I_{E2}=\frac{V_{E2}}{R_{E2}}$.



To maximize output range, as a starting point, we can pick a bias level at which $i_{E2}=I_{E2}+i_{e2}\sin\omega t$, where I_{E2} is the DC bias current at the emitter and i_{e2} is the AC component. To ensure that at the minimum emitter voltage the transistor stays out of cutoff with a minimum current of 20% of the DC bias current we can set $i_{e2}=0.8\,I_{E2}$. If you need to improve linearity you can adjust the factor of 0.8 lower to achieve a larger bias current and to stay further from cutoff at the signal minimum. Conversely if you need more power getting to the speaker you can push operation closer to cutoff by increasing this factor.

Looking out from the emitter, the AC component of this signal sees a resistance $R_L \vee ¿ R_{E2}$. At DC, the capacitor acts like an open circuit and so no DC current flows into the speaker and the resistance seen by the DC component is just R_{E2}. Thus the AC component of the emitter voltage is

$$v_{e2}=i_{e2}R_{E2}\vee ¿ R_L$$

Since the DC component of the emitter signal only sees R_{E2}, we also have

$$V_{E2}=I_{E2}R_{E2}$$

Since only the AC component drives the speaker,

$$i_L=\frac{v_{e2}}{R_L}=i_{e2}\frac{R_{E2}\vee ¿ R_L}{R_L}$$

So

$$i_{e2}=\frac{i_L R_L}{R_{E2}\vee R_L}$$

And if $i_{e2}=0.8\,I_{E2}$, then

$$0.8\,I_{E2}=\frac{0.8\,V_{E2}}{R_{E2}}=\frac{i_L R_L}{R_{E2}\vee R_L}$$

$$0.8\,V_{E2}R_{E2}R_L=i_L R_L R_{E2}\left(R_L+R_{E2}\right)$$

$$0.8\,V_{E2}R_L=i_L R_L\left(R_L+R_{E2}\right)$$

So that

$$R_{E2}=\frac{0.8\,V_{E2}-i_L R_L}{i_L}$$

Using $R_L=8\,\Omega$, $I_L=50\,mAp$ and $V_{E2}=2.15\,V$, we obtain $R_E=26.4\,\Omega$. With this value of R_E we have $I_{C2}\approx I_{E2}=\frac{2.15\,V}{26.4\,\Omega}=81.4\,mA$. $Q2$ therefore has a transconductance of $g_{m2}=\frac{81.4\,mA}{0.025\,V}=3258\,mA/V$!

Knowing g_{m2} allows us to calculate R_{i2} which will be important for designing the CE stage. According to the equation sheet it is $R_{i2}\approx r_\pi+\left(R_{E2}\vee R_L\right)\beta=1.289\,k\Omega$ (check this yourself!)

Designing the CE stage:

For the CE stage we need to simultaneously achieve a high gain and a high input resistance. It is helpful to rearrange the expressions for CE gain and input resistance to figure out the full implication of these requirements.

The input resistance will be limited by R_{i1} since $R_i<R_{i1}$ because R_{i1} appears in parallel with the base bias resistors at AC. Neglecting the effect of the base bias resistors for the moment, we have

$$R_{i1}<r_{\pi1}\left(1+g_m R_{E1}\right)$$

We can use this to rewrite the gain equation:

$$|G|=\frac{g_{m1}\left(R_{C1}\vee R_{i2}\right)}{1+g_m R_{E1}}$$

$$|G|=\frac{g_{m1}\left(R_{C1}\vee R_{i2}\right)}{R_{i1}}\times r_{\pi1}=\frac{g_{m1}\left(R_{C1}\vee R_{i2}\right)}{R_{i1}}\times\frac{\beta}{g_{m1}}=\frac{\beta\left(R_{C1}\vee R_{i2}\right)}{R_{i1}}$$

So that

$$|G|\times R_{i1}=\beta\left(R_{C1}\vee R_{i2}\right)$$

Here we can see how the relatively low input resistance of the second stage is going to limit us. Given the requirements we need $|G| \times R_{i1} > 15 \times 80\,k\,\Omega = 1200\,k\,\Omega$, but $(R_{C1} \vee |R_{i2}) < R_{i2}$ and $\beta R_{i2} = 258\,k\,\Omega$. We cannot meet the requirement with the given circuit topology! What do we do?

Darlington stage:

The fundamental issue we found in the last section is that the input resistance of the CC stage is limiting the gain-input resistance product $|G| \times R_{i1}$ in the CE stage. We'd like to add another stage between them that presents a high input resistance to prevent loading of the Q1 collector and boost the CE stage voltage gain. We can do this by adding another CC stage that can present a high impedance at the input and use its low output impedance to drive the base current into the final output CC stage. This is an example of voltage buffering as discussed in Lecture 7.

In fact, this extra CC stage can be added in a very elegant way. The new CC stage can have its emitter directly connected to the base of the original CC stage. This has the effect of increasing the input resistance seen looking into the second stage by a factor of β. This configuration of two cascaded CC stages is called the Darlington pair and was discussed in Lecture 10.

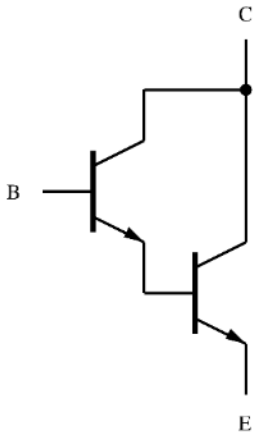
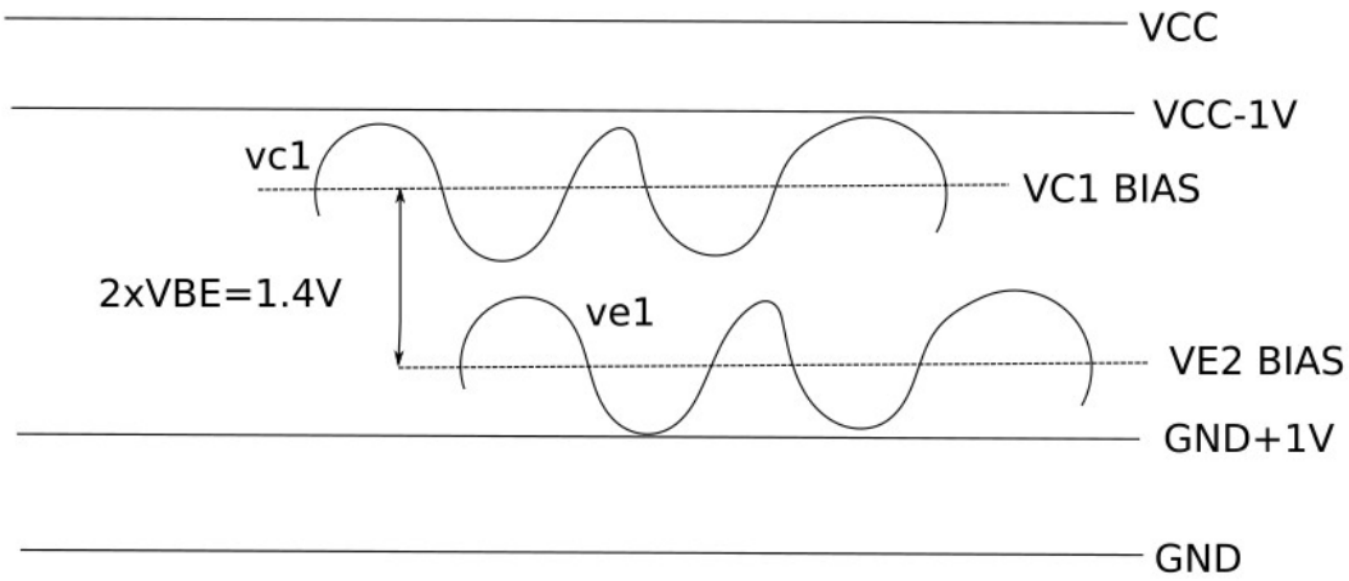


Figure 3: Two CC stages cascaded together to form a Darlington stage which has higher input impedance than a single CC stage

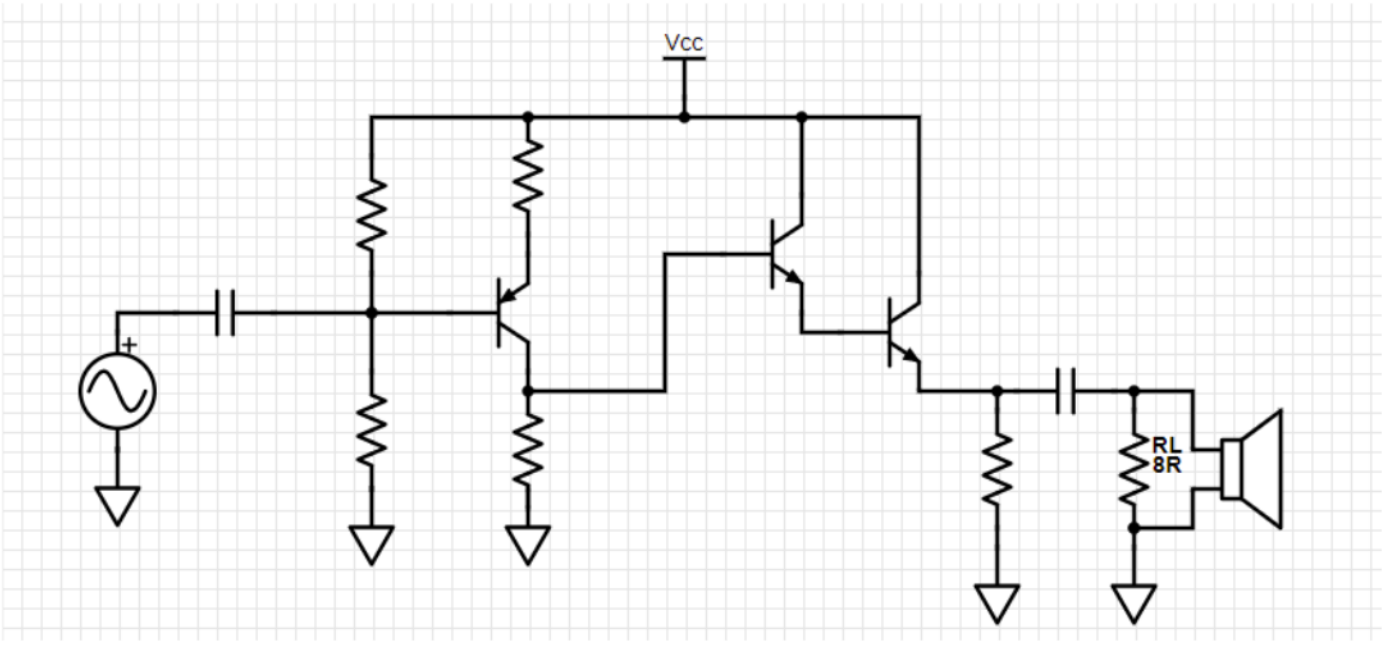
If you replace the original NPN transistor in your CC stage with a Darlington pair, the input resistance seen looking into the base increases by a factor of $\beta = 200$, i.e. from $1.2k\,\Omega$ to $247\,k\,\Omega$, making it possible to easily achieve $|G| \times R_{i1} = \beta(R_{C1} \vee |R_{i2}) > 1200\,k\,\Omega$ with appropriate choice of R_{C1}.

The thing you give up for this is head room as you now have two V_{BE} drops in the second stage instead of one drop.



This means that V_{E2} should now be $2.5\,V - 0.7\,V = 1.8\,V$ and V_{C1} should be $2.5\,V + 0.7\,V = 3.2\,V$, but this still allows you more than enough signal swing ($\pm 800\,mV$) to meet the requirements.

With the Darlington pair included in the circuit your final design should look like this:



The rest of the design

With two 2N2222s in Darlington pair configuration in the second stage, the first stage is almost completely decoupled from the second stage, assuming you choose $R_{C1} \ll R_{i2}$ $268\,k\,\Omega$. In other words, the gain of the first stage is determined almost entirely by your choice of R_{C1} and does not depend on anything going on in the CC circuit. This

allows you to design the first stage while treating $R_{i2} \approx \infty$ and just worry about meeting $A_v > 15$ and $R_i > 80\,k\,\Omega$ in the first stage. The calculations involved in this design are similar to what we did in Lab 4 and so you can apply the techniques you learned there to complete the design.

Dissipated power

The power dissipated in a BJT transistor is the time-averaged product of the voltage drop between the collector and emitter and the collector current $P = \langle v_{CE} \times i_C \rangle$. This power gets turned into heat in the transistor and will cause it to heat up. If $v_{CE} = V_{CE} + v_{ce}$ and $i_C = I_C + i_c$ are made up of a DC and a sinusoidal AC component then we can break these out as $P = V_{CE} I_C + \frac{1}{2} v_{ce} i_c$. For power amplifiers like this one, it is important to ensure that the dissipation of the output transistor (i.e. the transistor directly connected to the load) stays within a specified range to avoid thermal failure. The output current that a transistor can handle can often be greatly increased by careful heatsinking – attaching the transistor to a metal block that can suck away heat and/or by providing active cooling with fans or water cooling.

5 Simulating

I recommend using transient analysis for debugging and to ensure that transistors stay in the active region. Input impedance (R6-03) and power (R6-04) can be verified with an AC analysis.

Plotting power to verify R6-04 requires that you create a trace that multiplies load current by load voltage, and LTSpice will plot it in mW. Note that AC analysis in LTSpice reports current and voltage using their peak values rather than their RMS values, so you need to divide by 2 to get the power. In other words,

$$P = \frac{V_p \times I_p}{2}$$

Where P is power and V_p and I_p are the peak voltage and current obtained through AC analysis.

THD (R6-06) can be analyzed the same way as in Lab 4 using the **.fourier** Spice directive.

For requirement R4-05, it is adequate to show that the output stage transistor you used can handle the current and power. You can do this by comparing the specified maximum collector current $\overline{I_C}$ from the datasheet for the 2N2222 with the maximum current measured in your circuit in a transient simulation under the specified operating conditions. The power specified in the datasheet refers to the average dissipated power in the transistor $\overline{v_{CE} i_C}$. You can use a 1KHz sine wave for the input signal in this simulation. You can calculate the time-average of a signal in LTSpice by CTRL-Left Clicking on the name of the signal in the title bar.

6 Design Verification

In this lab you are expected to plan and execute the requirements on your own using the hints provided in the previous section. Refer to the example data posted for Lab 1 as a guide on how to format your test reporting. A template test is provided below.

Test T3-xx	

Lab 5: Multistage amplifiers

Requirement(s) being tested	
Output files	
Description of simulation configuration	
Description of test output	
Acceptance criterion	
Test result	
Objective evidence	